

FINAL EXPLANATION: BIOLOGY GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: BIOLOGY GRADE 10 – SUB-STRAND 2.2 PLANT TRANSPORT: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the unit. Use everything you learned in Lessons 1 to 8 to answer the driving question: How does water and food travel through an entire plant, from the soil to the highest leaf, and why does this journey sometimes fail, causing crops like maize in Kibwezi to wilt and die? Remember the Kibwezi farmer: she watered her maize in the morning, yet it wilted in the afternoon. Maize near the dry seasonal riverbed wilted sooner than maize on the wetter lower slopes, and sukuma wiki shaded by a mango tree barely wilted at all. Answer each section in full sentences in the space under each prompt. Use the key words you learned, such as xylem, phloem, osmosis, active transport, transpiration, cohesion-tension, turgor and pressure-flow. Where a section asks for a Claim-Evidence-Reasoning (CER) answer, give your claim, then your evidence from our investigations, then your reasoning. You may add a labelled diagram if it helps. Write in your own words.

Section 1: Why a plant needs a transport system

The Kibwezi farmer says, 'The soil has water, I can see it, so why is my maize wilting?' (a) Where do water and mineral salts enter a maize plant, and where does carbon dioxide enter? (b) Where in the plant is the water needed? (c) Use your answers to (a) and (b) to explain the farmer's puzzle. Name the two transport tissues that connect the entry points to the places of use.

(a) Water and mineral salts enter at the roots, below the soil. Carbon dioxide is taken in at the leaves, above the soil. (b) Water is needed in the leaves (for example for photosynthesis and to keep the cells firm), as well as in all other parts of the plant. (c) The place where water enters (the root) is far from the place where it is needed (the leaf). Soil water and leaf water are separated by the whole length of the plant. So the plant needs an internal transport system to carry water from the root to the leaf. If this system cannot deliver water fast enough, the leaf wilts even though the soil is damp. The two transport tissues are xylem, which carries water and dissolved mineral salts upward, and phloem, which carries manufactured food (sugars) from the leaves to other parts of the plant.

Section 2: Entry of water and mineral salts at the root

Describe the path taken by water from the soil to the xylem in a maize root, naming the structures it passes through. (a) What process moves the water, and what makes it move in that direction? (b) How do mineral

The water enters through root hair cells, crosses the cortex, passes through the endodermis (where the Casparian strip forces it through living cells) and enters the xylem. (a) Water moves by osmosis, passively, down a water potential gradient: it moves from where water potential is higher (the soil) to where it is lower (the root hair cells, then across the cortex and endodermis towards the xylem). Root hairs have a large surface area so they absorb quickly. (b) Mineral salts enter by active transport, moving against their concentration gradient, which needs energy (ATP) from respiration. Water uses no energy; salts do. (c) The potato strips modelled root hair cells in wet soil (tap water) versus drier soil with low water potential (concentrated salt solution). In tap water the strip gained 3 mm (50 mm to 53 mm) and stayed firm, because water moved in by osmosis. In the concentrated salt solution the

salts enter the root, and how is this different from water? (c) In your potato strip investigation, you used tap water and a concentrated salt solution. Each potato strip started 50 mm long. After 15 minutes the strip in tap water measured 53 mm and the strip in concentrated salt solution measured 46 mm.

Solution	Start length (mm)	Length after 15 min (mm)
Tap water	50	53
Concentrated salt solution	50	46

Use these results to explain what the potato strips modelled for the root, and how this can help explain why maize in drier soil near the seasonal riverbed wilts sooner than maize on the wetter lower slopes.

strip lost 4 mm (50 mm to 46 mm) and became soft, because water moved out. In the same way, when the soil near the dry riverbed is drier, its water potential falls below that of the root hair cells. Osmosis slows or stops, or water even leaves the root, so the plant cannot take in enough water and wilts sooner than maize on the wetter lower slopes.

Section 3: The journey up the stem and the loss at the leaf

Using Claim-Evidence-Reasoning, answer: Why does the Kibwezi maize wilt in the afternoon even though the farmer watered the soil that morning? In your Evidence, use what you observed in our leaf investigations and this class

Claim: The maize wilts because on a hot, dry, windy afternoon the plant loses water by transpiration faster than the roots can absorb it and the xylem can replace it, so the leaf cells lose turgor. Evidence: In the sweating leaf bag, water droplets collected inside the bag, showing that leaves release water vapour. The stomatal impressions from sukuma wiki showed stomata with guard cells, with more on the lower epidermis, which are the exits for the vapour. In the potometer, the uptake rate was 6 mm/min in the control, but 15 mm/min with wind and 13 mm/min with the lamp, so wind and heat more than doubled water uptake. When petroleum jelly blocked the stomata, the rate fell to 2 mm/min, much lower than the control. This shows that transpiration through the stomata drives how fast water is taken up. Reasoning: Water evaporates from the leaf mesophyll cells and leaves through the stomata (transpiration). This lowers the water potential in the leaf and creates tension (transpiration pull) that pulls the water

potometer data (a potometer measures how fast a cut leafy stem takes up water; the air bubble moved along the capillary tube for 2 minutes in each condition).

Condi on	Bubble distanc e in 2 min (mm)	Uptake rate (mm/m in)
Control (still air)	12	6
Wind (fan)	30	15
Heat (lamp)	26	13
Petrole um jelly on leaves	4	2

In your Reasoning, explain how water is pulled up from the roots to the leaves, and link the conditions that change the rate of water loss to a Kibwezi afternoon.

column up the xylem. Water molecules stick together (cohesion), so the whole column is pulled up from the roots; root pressure gives a small extra push. Four factors increase the transpiration rate: high temperature, high wind speed, low humidity and high light intensity. All four occur on a dry Kibwezi afternoon. When the loss is greater than the supply, the leaf cells lose turgor and the plant wilts.

Section 4: Why some plants recover, some die, and shade protects sukuma wiki

The farmer noticed that by evening some maize plants recovered while others did not, and that the sukuma wiki under the mango tree barely wilted. (a) Explain why shade protects the

(a) The mango tree's shade lowers the light intensity and the temperature around the sukuma wiki leaves, which are two of the four factors that affect transpiration. The vapour pressure gradient between leaf and air is smaller, so the stomata need not open as wide and less water is lost. The xylem supply can keep up, turgor is maintained, and the plant does not wilt. (b) When transpiration is very high and the water supply is low, the tension in the xylem becomes very large. Air can enter the narrow xylem vessels and break the water column (an air embolism, or cavitation). The water highway is then broken, so water cannot reach the leaves even when the farmer waters again, and the plant dies. Plants that recover were able to keep their water columns unbroken and regain

sukuma wiki, naming which transpiration factors the shade changes. (b) Explain why some maize plants never recover, even after the farmer waters the soil again. Think about what could happen inside the stem when water loss is very high and supply is low. (c) Why do the shaded sukuma wiki plants have a lower risk of this problem?	turgor in the cooler evening. (c) Shaded sukuma wiki loses less water, so the tension in its xylem stays lower and the water columns are much less likely to break.
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Section 5: The return journey of food and the complete model	
Plants also move food. (a) Which tissue carries sucrose and amino acids, from where to where, and what did the girdled eucalyptus trees near Kericho show you about this? (b) Explain the pressure-flow hypothesis in your own words. (c) Finally, summarise the whole plant transport journey in order, from water entering a root hair to sugar reaching a sink such as the maize cob, and say how this journey can fail in the Kibwezi maize.	(a) Phloem sieve tubes, helped by companion cells, carry sucrose and amino acids from source cells (photosynthesising leaves) to sink cells (roots, fruits such as the maize cob, and growing tips). In the girdled eucalyptus trees near Kericho, where a ring of bark (phloem) was removed, sugars built up above the girdle and the roots below starved and died. This is direct evidence that phloem carries sugars downward from the leaves. (b) Sucrose is loaded into the phloem at the source, which makes the sieve tubes there very concentrated. Water enters by osmosis and builds up high pressure. At the sink, sucrose is removed and used, water leaves by osmosis, and the pressure is lower. The sap is pushed by mass flow from the high-pressure source to the low-pressure sink. (c) Water enters root hair cells by osmosis down a water potential gradient, crosses the cortex and endodermis and enters the xylem. Mineral salts enter by active transport. Transpiration at the stomata creates tension, and cohesion-tension pulls the water column up the xylem to the leaves, with root pressure helping a little. In the leaves, sugars are made by photosynthesis, loaded into the phloem and moved by pressure-flow to sinks such as roots and the maize cob. The journey fails when transpiration is greater than uptake (hot, dry, windy afternoons, or low-water-potential soil near the riverbed). Turgor is lost and the maize wilts, and if the xylem columns break the plant cannot recover, and food transport is also disrupted.

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Scientific accuracy of the transport pathways	Correctly describes water and mineral entry (osmosis, active transport), upward flow in xylem and sugar flow in phloem, using the correct terms throughout.	Describes most pathways correctly with minor errors or missing terms (for example confuses one direction or omits the endodermis).	Confuses xylem and phloem, or the processes, or gives incomplete pathways.
Use of evidence from the investigations	Cites specific evidence and numbers from our lessons (potato strips, sweating leaf bag, stomatal impressions, potometer, girdled eucalyptus trees near	Cites some evidence from the investigations but links it to claims only partly.	Gives little or no evidence, or only general statements.

	Kericho) and links each piece to a claim.		
Explaining the Kibwezi phenomenon	Fully explains why maize wilts despite watered soil, why riverbed maize wilts sooner, why some plants die and why shaded sukuma wiki is protected, using rate of loss versus supply.	Explains most parts of the phenomenon but misses one (for example the shade or the riverbed explanation).	Explains only that the plant needs water without linking to transpiration or soil water potential.
Reasoning with mechanisms (CER, cohesion-tension, pressure-flow)	Gives clear, logical reasoning connecting cause and effect, including cohesion-tension, the four transpiration factors, and pressure-flow.	Gives reasoning that is mostly logical but with gaps in one mechanism.	States facts without explaining how or why.
Communication and Kenyan context	Writes clearly and in order, using the key terms correctly and linking ideas to the Kibwezi farm context.	Writes understandably with a few unclear parts or terms used loosely.	Writing is hard to follow and key terms are missing or misused.

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Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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